When Test-Time Search Exploits Latent Costs in Robot World-Model Planning

Anonymous*

Abstract

Action-conditioned JEPA world models plan by coupling predicted latent futures to a representation-induced goal cost and a test-time optimizer. Published systems succeed at reaching and pose control, but remain brittle on pushing, grasping, and pick-and-place. We localize one failure at the optimizer–cost interface. In MetaWorld, an *oracle ladder* rolls every candidate through the simulator and re-encodes the resulting observation, removing learned dynamics error while varying the cost. With the same planning budget, a privileged true-state cost solves push 63/64 and pick-place 41/64; latent L_2 solves 0/64 on both tasks, and a probe implementation of the same state-cost form reaches only 4/64 and 0/64 on DINO-WM and 1/64 and 0/64 on JEPA-WM. On identical first-iteration populations, probe-selected elites have 1.24–1.40 cm larger object-readout error than the population and incur 2.05–2.47 cm physical opportunity regret. A matched-residual permutation null preserves the same error magnitudes but explains only 1.07–1.15 cm of regret, showing structured residual alignment beyond a generic noisy-score winner's curse. A paired branch intervention starts from identical candidates and changes only the cost used to refit CEM; after five refits, the probe branch's best available physical cost is 1.51–2.01 cm worse across both checkpoints and tasks. Exact task-success coverage is nevertheless sparse in contact populations, so the result identifies both proposal coverage and cost ranking as bottlenecks rather than attributing all failure to selection. We contribute an optimizer-conditioned, simulator-verified audit of cost reliability under perfect dynamics. The claim is scoped to the tested MetaWorld tasks, checkpoints, costs, and planners; it does not establish that the encoder alone is defective or that every representation-derived cost must fail.

1 Introduction

Action-conditioned world models that predict in a learned latent space — the JEPA family (DINO-WM (Zhou et al., 2024), V-JEPA-2-AC (Assran et al., 2025), and the design study of Terver et al. (2025)) — plan by coupling two learned components with a test-time optimizer: a predictor imagines futures, a cost scores them against a goal, and a sampling planner (CEM) searches for the action sequence that minimizes the cost. On free-space tasks this works well. But a consistent pattern runs through the field's own evaluations: the *positive* closed-loop numbers are for reaching and pose control, while contact-rich tasks — pushing, grasping, pick-and-place — are omitted, demoted to qualitative figures, or propped up with hand-crafted sub-goals. Terver et al. (2025) stop their closed-loop Metaworld tables at Reach and note the model "hallucinates grasping the object"; Assran et al. (2025) report pick-and-place only after supplying three manual sub-goal images.

One common diagnosis is a *model-capacity* story: the encoder is too small, the representation too coarse, or the predictor too weak. We isolate a more specific failure: the **optimizer–cost interface**. A representation can be predictive while exposing an inaccurate terminal cost, and a sampling planner acts on candidate *rankings*, not average reconstruction error. Optimizing a noisy proxy is already known to produce Goodhart or optimizer's curse effects (Gao et al., 2023; Skalse et al., 2022; Karwowski et al., 2024). Our question is therefore not whether noisy costs can mislead optimization in principle. It is whether a released robot-planning stack fails after learned dynamics are removed, whether simulator truth can locate the ranking failure on candidates actually considered, and whether the cost used for adaptive refitting changes the physical quality of later populations.

*Draft v0.5, 2026-07-16. Confirmatory results are included only when marked paper-ready in `diagnosis/docs/CLAIMS_EVIDENCE.md`.

This diagnosis is not new; the isolation and the survival test are. A concurrent cluster of work, built mostly on small, non-robot latent world models, converges on a compatible reading — predictive accuracy does not imply a plannable cost. We do not claim first discovery that cost geometry, not the predictor, is the bottleneck (Sec. 2 compares each account in detail). What we add is a controlled robot-planning test: an oracle ladder that excludes learned-dynamics error, a same-population audit that scores every candidate by both proxy and simulator truth before selection, and a shared-population branch that changes only the cost used for adaptive refitting. The main evidence uses MetaWorld, where simulator state provides an independent verifier. The DROID diagnostic is observational supporting context, not causal transfer evidence.

Three features distinguish our account from the classic "model exploitation" concern in model-based RL (Janner et al., 2019; Lambert et al., 2020), where a policy exploits errors in learned *dynamics*. (i) We give the planner *perfect dynamics* — every candidate action sequence is rolled in the simulator and re-encoded by the real frozen encoder — and the failure persists. Learned-dynamics error is therefore excluded within this harness, localizing the tested failure to cost and search. (ii) The privileged cost is a positive control: it demonstrates that the same finite search budget can solve the physical task. (iii) The probe cost is already imperfect in task-relevant relative geometry, and adaptive selection further concentrates favorable residuals. We do not present this as a globally accurate encoder mysteriously failing; we measure how an imperfect representation–readout cost becomes a physical ranking error.

Concretely, our experimental chain (MetaWorld with simulator ground truth for isolation and verification; DROID as an observational real-robot transfer check) supports three contributions:

- **Observational diagnosis across released checkpoints (Sec. 3).** Against dataset-mined, nearby-state action alternatives, four frozen DROID checkpoints from 22M to 1B parameters show no improvement in factual-action ranking. MetaWorld exhibits the same pre-grasp-worst ordering. Because the alternatives originate from neighboring rather than identical states, this is a discriminability diagnostic, not a causal intervention or controlled scaling law; a snapshot/restore benchmark is the required validation.

- **Oracle localization under locked evaluation (Secs. 4–6).** Our harness first matches the released Reach result within uncertainty and reproduces the contact wall. On 64 unseen paired seeds, true-state cost solves push 63/64 and pick 41/64, while latent L_2 and probe costs remain near zero across DINO-WM and JEPA-WM. Learned dynamics error is therefore not necessary for failure in this harness.

- **An optimizer-conditioned cost audit (Sec. 8).** We separate average-case readout quality from same-population misranking, opportunity regret, proposal coverage, and the effect of branch-specific refitting. Probe elites have CI-clean excess error and regret on identical candidates; after refitting, their branch contains worse best physical candidates. Coverage is also limited, so neither a selection-only nor a search-only explanation fits every cell.

Conventions (how to read our numbers). Four kinds of quantities appear throughout, and we keep their formats distinct. *Planning success* is episodes solved out of the stated paired evaluation set; headline oracle cells use 64 unseen seeds, while exploratory and ablation cells use 8 or 16 as marked. *Readout accuracy* is the percentage of frames whose decoded object position falls within the 5 cm task-success radius ("% within 5 cm"), with the median error in cm where informative. *CRA* is a ranking accuracy over transitions, reported as a fraction in tables; with 16 exchangeable distractors the nominal baseline is $1/17 \approx 0.059$, but mined neighbours need not be exchangeable. *AUG/ECS* are raw latent-space gaps (Sec. 3.1), meaningful only within a model. Distances in the workspace are in cm or in Metaworld state units where noted.

Planner conventions. We write a CEM budget as (samples)×(iterations) evaluated per replan — e.g. 100×6 is 100 candidate action sequences refined over 6 CEM iterations — with the rollout horizon H stated separately. At each iteration CEM keeps the top-scoring candidates, the *elites*, and refits its sampling distribution to them; the converged elite population is the set of futures the planner "trusts". Pools marked "n_{eff}-weighted" average per-regime cells weighted by their effective (trajectory-clustered) sample count, so a few long trajectories do not dominate.

The through-line is not that all static metrics are strong. Object-only localization can look adequate under a coarse threshold while end-effector and hand–object geometry remain poor. The planning-relevant signal is whether the *composite cost preserves physical candidate ordering* on the proposal distribution induced by the optimizer. We propose reporting this ordering, opportunity regret, and candidate coverage alongside end-to-end success.

2 Related work

JEPA world models for planning. DINO-WM (Zhou et al., 2024) predicts forward in the frozen feature space of a self-supervised image encoder and plans with sampling-based MPC from a goal image; V-JEPA-2-AC (Assran et al., 2025) learns an action-conditioned world model on a video encoder and demonstrates zero-shot manipulation; Terver et al. (2025) systematically study the design choices that make such models plan. We take these as frozen baselines and treat their own evaluations as evidence: across all three, strong closed-loop results are pose/free-space tasks, and contact is where the story thins (hand-crafted subgoals, qualitative figures, "hallucinated grasping"). V-JEPA 2.1 (Mur-Labadia et al., 2026) subsequently improves dense spatial features and reports a large real-robot grasping gain, making representation quality and observability an important alternative explanation. Our study does not rule that out; it asks whether the costs induced by the released planning checkpoints survive their optimizer's own selection.

Overoptimization and reward hacking. Optimizing against a learned proxy eventually decouples proxy from truth: formalized for RLHF reward models by Gao et al. (2023), characterized in general by Skalse et al. (2022) and Karwowski et al. (2024), and anticipated as "reward hacking" (Amodei et al., 2016). We use the same mathematical lens without claiming the general phenomenon as new: CEM optimizes a proxy terminal cost, and finite-population selection can concentrate favorable residuals. Our contribution is the controlled measurement available in simulation: dynamics can be held perfect, every candidate can be scored by physical truth, and the cost used for adaptive refitting can be intervened on.

Model exploitation in model-based RL. That policy optimization exploits errors in learned *dynamics* is a classic MBRL concern (Janner et al., 2019), running through the sampling-MPC lineage our CEM planner inherits (Chua et al., 2018; Nagabandi et al., 2019), related to the objective mismatch between likelihood training and control performance (Lambert et al., 2020), and addressed offline with pessimism (Yu et al., 2020b; Kidambi et al., 2020); learned agents more broadly admit adversarial exploitation of their blind spots (Gleave et al., 2020). Concurrently, Parthasarathy et al. (2025) show *gradient-based* planners exploit a world model's input gradients and respond by smoothing the dynamics. Our finding is materially different from all of these: with dynamics oracle-controlled — the channel every one of those works attacks or repairs closed entirely — the *cost* is exploited at test time, and the standard dynamics-side remedies (ensembles, pessimism) do not touch it (Sec. 6).

Action-conditioned reliability diagnostics. Recent action-verification work separates state plausibility from action reachability and uses inverse dynamics as a verifier (Liu et al., 2026); ATM similarly probes action information in real and predicted latent transitions (Chen et al., 2026). These questions motivated our earlier diagnostic, but action-identifiability is not a novelty claim of this paper. Our main target is candidate-cost ranking under simulator-controlled dynamics.

Closest cost and planner-interface diagnoses. Several concurrent arXiv preprints built on LeWorld-Model (Maes et al., 2026) reach compatible diagnoses by different routes; we treat them as important concurrent work, not accepted venue results. RC-aux (Li et al., 2026b) makes the framing explicit — "predictive but not plannable": accurate short-horizon prediction can still leave a latent whose Euclidean distance does not reflect finite-horizon reachability — and repairs it with a training-time reachability objective that reshapes the representation (nearer to our relearned-representation arm, which also fails to cross contact, than to a purely post-hoc metric). Trajectory reachability metrics (TRM) (Li et al., 2026a) instead add a post-hoc terminal-ranking head on the frozen backbone. TRM is the closest cost-side study: it deploys the repaired metric inside CEM, audits ordering on identical candidates, compares against true-state cost, intervenes on the task-relevant subspace, and stress-tests larger search budgets. ACID (Seo et al., 2026) argues, in the same spirit, that a terminal-only latent cost cannot see whether intermediate transitions are realizable, and adds an inverse-dynamics cycle-consistency term to the planning cost. IMWM (Gao et al., 2026) replaces learned dynamics with literal environment rollout and finds many failures arise because no successful candidate enters the finite CEM population. Thus neither latent-cost misranking nor oracle-dynamics failure alone is our novelty.

Our narrower target is a contact-rich robot regime in which we jointly vary dynamics and cost, inspect optimizer-selected elites in simulator object state, and measure candidate coverage and cost-based selection separately. Relative to IMWM, contact populations in our harness are often coverage-limited as well; when physical alternatives are present, opportunity regret additionally exposes misranking. Relative to TRM, our question is whether a learned terminal ranking survives fresh adaptive search on held-out MetaWorld episodes. Direct TRM- and ACID-style comparisons are ongoing and are required for a broad mitigation claim.

Amortizing away the search. Closest to our "remove the adversary" control (Sec. 8), Nguyen et al. (2026) replace CEM entirely with a goal-conditioned inverse-dynamics map and argue the opposite thesis — that a well-regularized latent has "already encoded" the structure search would recover, so the planner is not the bottleneck. Their evidence carries our wedge: the single contact-rich cell in their toy suite (Push-T) is the one place amortization stops dominating, which they attribute to "delicate multi-step contact." Our MetaWorld E1 similarly finds that the tested amortized controller does not move the object. It localizes a second failure outside search without deciding whether the cause is representation, controller training, or coverage.

Reconciling coverage, ranking, and search. The concurrent accounts need not be mutually exclusive. IMWM identifies a finite-query coverage bottleneck and adds demonstration priors; Nguyen et al. (2026) remove sampling search with an amortized controller; TRM repairs terminal ranking. Our experiments measure all three axes in one contact harness. Exact successful candidates are sparse, so coverage matters. Yet on identical populations the proxy-selected candidate also has positive physical opportunity regret, and branch-specific refitting worsens the best physical candidate subsequently available. The supported conclusion is therefore a joint coverage-and-ranking bottleneck, not a universal claim that search itself is either necessary or harmful.

Benchmarks. We evaluate on Metaworld (Yu et al., 2020a), whose simulator state lets us supervise and *verify* costs against ground truth, and DROID (Khazatsky et al., 2024) as the real-robot transfer check. Planning follows the upstream CEM configuration (Rubinstein, 1997).

3 Observational diagnosis: weak action discrimination near contact

We first measure how well frozen predictors discriminate the factual action from actions mined at nearby dataset states. These are observational, near-state queries: they do not provide the true successor of an alternative action from the identical simulator state. The diagnostic can localize a failure mode, but causal action-effect claims require the snapshot/restore intervention described in Sec. 8. Full definitions and the AUG/ECS corroboration are in App. B and App. F.

3.1 Metrics

Let $F(z, a)$ be the frozen one-step latent predictor (driven through the model's own `unroll` primitive, with the model's own action normalisation) and d the upstream planning distance (L_2 for every baseline, matching the shipped `L2_cem` configuration). We measure three complementary quantities on held-out factual transitions (z_t, a_t, z_{t+1}); each answers a different question.

CRA — does the model rank the executed action best? Given the transition, we sample K=16 distractor actions $\{a_k^-\}$ and ask whether the factual action's prediction lands closer to the realised next latent than every distractor's prediction:

$$\text{CRA} = \Pr\Big[\, d\big(F(z_t, a_t), z_{t+1}\big) < \min_{k=1\ldots K} d\big(F(z_t, a_k^-), z_{t+1}\big) \Big], \tag{1}$$

The value $1/(K+1) = 1/17 \approx 0.059$ is the nominal baseline only when factual and distractor actions are exchangeable; the dataset-mined sampler below is not exchangeable, so an empirical action-ignoring null is required for a calibrated chance claim. The distractor family controls what the metric tests: *opposite* distractors ($a^- = -a_t$) test coarse directional grounding — the quantitative version of the "open vs. close the gripper" visualisations in prior work. For the sampler called *hard-NN*, we first retain candidate transitions within a latent radius (falling back to the K nearest states when the radius is under-filled) and then select the K actions farthest from the factual action. Thus it is a stress test of factual-action retrieval under a model-specific neighborhood, not an intervention and not automatically comparable across models. We report it because it motivated the oracle study, not as the paper's primary causal evidence.

AUG — does the action help at all? The Action Usage Gap is the prediction-error increase when the action is replaced by another action from the batch (a permutation π):

$$\text{AUG} = \mathbb{E}\big[\, d\big(F(z_t, \pi(a_t)), z_{t+1}\big) - d\big(F(z_t, a_t), z_{t+1}\big) \,\big], \tag{2}$$

which is ≈ 0 when shuffling does not change average error. This is consistent with action-ignoring but is not an if-and-only-if test: signed differences may cancel and shuffled actions may be off-support. AUG is a raw latent-space gap: informative *within* a model, not comparable *across* models (latent scales differ), which is why it corroborates CRA rather than replacing it (App. F).

ECS — the same question, asked where it matters. Averaged over all transitions, AUG can be diluted by near-static steps (the arm hovering; the scene unchanged) where different actions genuinely lead to similar futures. Effect-Conditional Sensitivity restricts AUG — and, in the *effect-conditioned CRA* we report as the primary signal, the ranking — to transitions that actually move the latent, $\|z_{t+1} - z_t\| > \tau$ with τ the per-model median displacement. This removes many quiet transitions, but it also selects a different latent-defined subset for each model and can include ego-motion or nuisance variation. Cross-model claims therefore need a shared, model-independent physical-effect mask.

Boundary Blindness (BB; definition in App. B) is our fourth instrument. On a neighborhood of similar but non-identical states, it compares dispersion of observed outcomes with dispersion of model predictions under the neighbors' actions, both z-scored over the population; BB > 0 indicates an observational sensitivity shortfall. Initial-state, policy, task, and hidden-contact differences can also increase the observed fan. On DROID the outcome itself is a $\|\Delta z\|$ proxy, not independent world ground truth.

Protocol. Each transition is assigned to one of four mutually exclusive regimes — free space, pre-grasp, gripper actuation, contact manipulation — by a fixed-threshold priority cascade on the simulator state: gripper actuation if the gripper aperture changes, else contact manipulation if the object moves (> 5 mm), else pre-grasp if the hand is within 10 cm of the object, else free space (full thresholds, and the DROID gripper-sensor and RoboCasa grasp-flag analogues, in App. A). This is an object-displacement *proxy*, not a MuJoCo contact sensor — the released data ships no contact flags — and the caveat is carried on every figure. All confidence intervals are trajectory-clustered bootstrap. Every frame is encoded once and cached; nothing is trained for the diagnostic.

3.2 Frozen baselines fail on nearby actions, at the boundary

The observational Boundary-Blindness score localizes the same pattern: pooled $\text{BB}_{\text{boundary}}$ concentrates at the pre-grasp boundary on both model families (DINO-WM 1.323, JEPA-WM 1.280, vs. free-space 0.28–0.30; per-task CI-aware pairing elevated in 4/6 and 5/6 tasks with zero confident reversals) and transfers to real-robot DROID data (DINO-WM 1.916 $[1.560, 2.279]$ under the $\|\Delta z\|$ proxy; Table 2 gives the full four-model breakdown). A conditional-mean account is compatible with this pattern when observations alias multimodal contact outcomes, but the observational neighborhood alone does not identify that mechanism.

3.3 No upward trend across four released checkpoints

We score four frozen DROID world models on an identical 333-episode subset: DINO-WM (DINOv2 ViT-S, 22M), JEPA-WM (DINOv3 ViT-L, 300M), V-JEPA-2-AC (ViT-G, 1.01B), and the independent public

Table 1: Effect-conditioned factual-action ranking accuracy (top-1, n_{eff}-weighted pool, Metaworld). Against *opposite* distractors the models are near-perfect — their predictions respond to coarse action changes. Against the hard-NN stress-test they score lower, worst at pre-grasp. Because hard-NN is near-state and action-maximizing, this table motivates rather than proves a same-state contact bifurcation.

model	distractor	free space	**pre-grasp**	contact
DINO-WM	opposite	0.990	0.961	0.975
	nearest-neighbour	0.566	**0.467**	0.481
JEPA-WM	opposite	0.993	0.972	0.984
	nearest-neighbour	0.635	**0.541**	0.571

V-JEPA-2-AC-OSS (1B).

Table 2: Model-native DROID action-discriminability diagnostic across four released checkpoints. *Top:* effect-conditioned CRA against nearest-neighbour distractors (nominal exchangeable-candidate baseline $1/17 \approx 0.059$): all four models fall in a narrow range in the action-critical regimes (0.031–0.077). Because the effect mask and nearest neighbours are defined in each model's own latent geometry, this is an observational comparison rather than a controlled scaling law. *Bottom:* $\text{BB}_{\text{boundary}} > 0$ in every cell (boundary-blind), pre-grasp locus in 3/4. A positive control rules out metric saturation: on toy fully-actuated datasets the same diagnostic returns 0.66–1.0.

regime	DINO-WM 22M	JEPA-WM 300M	V-JEPA2-AC 1B	V-JEPA2-AC-OSS 1B
effect-CRA, nearest-neighbour (chance ≈ 0.059*)*				
pre-grasp	0.068	0.077	0.061	0.069
gripper actuation	0.031	0.059	0.049	0.031
contact manipulation	0.055	0.045	0.035	0.033
$\text{BB}_{\text{boundary}}$ *(> 0 = boundary-blind)*				
pre-grasp	**1.916**	**1.205**	**1.933**	**1.560**
free space	0.963	1.103	1.010	0.819
gripper actuation	1.088	1.057	0.643	0.828
contact manipulation	0.421	0.897	0.390	0.595

The model-native scores show no upward trend across the released checkpoints. They do not establish that parameter scale has no effect: each model supplies its own effect mask and nearest-neighbour set, and the candidates are not exchangeable interventions. A preregistered shared physical effect mask with fixed negative indices is therefore required for a controlled scaling claim. Nor does this section establish that weak action discrimination causes planning failure. Sections 5 and 6 instead test the planner–cost interface directly.

4 The wall, at upstream parity

We match the upstream Metaworld closed-loop evaluation read off its shipped configuration: goal = the scripted expert's final frame; CEM with horizon 6, 300 samples, 15 iterations, executing 3 model-steps per replan, $\leq$ 100 env steps; success = the simulator's flag at episode end. Reaching a faithful protocol required fixing three environment-side reproduction bugs, documented with verification probes in App. E (after the fixes, rendered observations are on the training manifold: one-step prediction error 1.4× dataset level, latent nearest-neighbour ratio 0.97 (a rendered frame's distance to its nearest dataset latent, relative to the dataset's own nearest-neighbour distance; ≈1 means indistinguishable from training frames); physics identical, 1.5 mm replay error).

Three readings. (i) Where the diagnostic is clean (free space), the harness is consistent with the published baseline: Reach 6/16, i.e. 37.5% with CI $[18.5, 61.4]$%, against the reported 44.8±8.9% — the published value

Table 3: Closed-loop Metaworld, 16 paired episodes per task (paired env initialisation and CEM noise; episode-end judging). The harness recovers a Reach rate consistent with the published baseline; contact collapses with the diagnostic signature. "grounded" = the L_2 cost augmented with a 0.5M object-dynamics channel $h(z, a) \to \Delta$obj (App. C).

task	arm	success	final state-dist	final ee-dist
mw-reach	L_2	**6/16 (37.5%)**	0.332	0.022
	grounded	**8/16 (50.0%)**	0.412	0.031
mw-push	L_2	0/16	0.624	0.038
	grounded	0/16	**0.527**	0.028
mw-pick-place	L_2	0/16	0.578	0.035
	grounded	0/16	**0.496**	0.021

falls inside our interval, though our point estimate sits below its mean; we read this as compatibility from one released checkpoint over 16 episodes, not a seed-matched replication (cf. limitations, Sec. 8). The harness is healthy, the contact failure is real. (ii) The failure signature is specific: final end-effector 2–4 cm from its goal pose while the object never moves — the closed-loop face of the smeared contact bifurcation, matching the upstream reports ("hallucinates grasping"; sub-goal images required). (iii) A grounded auxiliary cost (a 0.5M object-dynamics channel $h(z, a) \to \Delta$obj whose construction and metric-level gains we defer to App. C) improves the cost surface with a CI-clean paired gain (final state-dist +0.089 [+0.022, +0.162] pooled contact) — *and flips zero successes*. That last observation is the loose thread this paper pulls: if the model is the problem, why does repairing the model's outputs not repair planning?

5 The oracle ladder: the wall is the cost

We isolate which component breaks contact planning by swapping exactly one rung at a time, holding the planner fixed (CEM 100×6 per replan, H=6, strict episode-end success). Headline arms use 64 unseen paired seeds on both `dino_wm_metaworld` and `jepa_wm_metaworld`. This budget is smaller than the parity harness of Sec. 4 (300×15) for compute reasons, but it is *not* the bottleneck: Sec. 6.5 sweeps the CEM budget up to 24 iterations and 300 samples and the contact rungs never improve, while the same budget solves the task when supplied the informative true-state cost. The key device is the *latent oracle*: every CEM candidate is rolled in the *simulator*, rendered, and encoded by the real frozen encoder — perfect dynamics through the true representation, learned predictor removed. Any rung that fails under the latent oracle cannot attribute that failure to learned dynamics; within this harness the remaining tested bottleneck is the cost and its interface with search.

Table 4: Locked oracle ladder (64 unseen paired seeds; strict episode-end success). Every row uses the same CEM budget and simulator dynamics. The privileged cost demonstrates that the task and budget admit success; replacing simulator state by frozen-representation costs collapses both checkpoints.

rung	model	cost	push	pick
state oracle	—	true state	**63/64**	**41/64**
latent oracle	DINO-WM	latent L_2	0/64	0/64
latent oracle	DINO-WM	probe state cost	4/64	0/64
latent oracle	JEPA-WM	latent L_2	0/64	0/64
latent oracle	JEPA-WM	probe state cost	1/64	0/64

The ladder establishes two points. First, learned-dynamics error is not *necessary* for failure in this harness: every candidate endpoint is a true simulator future, yet the representation-derived costs remain near zero. Second, the planner implementation and finite budget are not by themselves an explanation: with privileged state cost, the same search succeeds on 98.4% of push and 64.1% of pick episodes. The remaining gap is at the tested cost interface. This does not prove that the encoder alone lacks information; the interface contains the encoder, the probes, their training distributions, and the composite cost.

Object-only decoding is not a certificate for the state cost. On random-action off-policy frames, a hardened object probe reaches median 2.01 cm error and 91.5% within 5 cm. We do not call the whole cost accurate on this basis. The end-effector probe has median 5.14 cm error and only 45.8% within 5 cm; the decoded hand–object relative vector has median 11.18 cm error and only 3.7% within 5 cm. Since the probe cost is $\|\hat{o}-\hat{g}\|+0.5\|\hat{h}-\hat{o}\|$, this relative geometry is directly task-relevant. Improving the object-only metric from 78.3% to 91.5% within 5 cm still yields only 1/16 push in the original re-gate. The supported conclusion is that coarse object localization alone is insufficient, not that an otherwise precise cost mysteriously fails.

6 Optimizer-conditioned cost misranking

6.1 Same-population selection before adaptive refitting

The oracle success gap alone says only that the probe cost is inaccurate. To test whether selection concentrates favorable errors, we inspect the first CEM population *before* any refit. Every one of the same 100 action sequences is rolled in MuJoCo and assigned both the stateprobe proxy $\hat{c}$ and the simulator-state cost c^*. We then compare the top 10% under $\hat{c}$, the top 10% under c^*, and the full population. Define opportunity regret

$$R_{\text{sel}} = c^*\Big(\arg\min_a \hat{c}(a)\Big) - \min_a c^*(a), \tag{3}$$

and optimism $o(a) = c^*(a) - \hat{c}(a)$. Both costs use the same object-plus-hand functional form and metre units; their difference is induced by decoded versus simulator state.

Table 5: First CEM population, 16 unseen seeds per cell. Object decode error is the mean in cm. Optimism enrichment compares proxy elites with the full population. All reported enrichments and regrets have seed-clustered 95% CIs strictly above zero.

cell	population	proxy elite	truth elite	optimism enrich.	R_{sel}
DINO push	5.18	6.45	5.50	2.33 cm	2.47 cm
DINO pick	4.98	6.23	5.17	2.16 cm	2.25 cm
JEPA push	4.79	6.03	5.12	2.28 cm	2.05 cm
JEPA pick	4.78	6.18	5.16	2.21 cm	2.06 cm

Proxy elites have 1.24–1.40 cm more object error than their own raw population and 0.91–1.06 cm more than simulator-truth elites on the same candidates. More importantly, their optimism is enriched by 2.16–2.33 cm and the selected action incurs 2.05–2.47 cm physical opportunity regret. Thus the low proxy ranking is partly produced by residuals with the favorable sign, rather than by physical progress alone. This is an optimizer-conditioned selection effect in the representation–readout–cost composition; it does not assign the error specifically to the encoder.

To distinguish structured misranking from the winner's curse that arises when minimizing any noisy score, we construct a matched-residual permutation null. Within each identical first-iteration population, we keep the simulator cost and the exact multiset of stateprobe residuals, but randomly permute residuals across candidates before selecting the minimum. The shuffled score therefore has the same marginal error distribution and search budget while breaking the association between residual sign and candidate geometry. Its argmin regret is 1.07–1.15 cm, whereas the real score incurs 2.05–2.48 cm; the actual-minus-null excess is 0.91–1.36 cm, with seed-clustered 95% CIs excluding zero in all four cells. Thus ordinary selection over a noisy cost explains part, but not all, of the observed regret.

6.2 Supporting distribution shift

The older random-reference comparison asks a different question: how the probe behaves on generic random-action frames versus CEM elites after selection. It supports distribution shift but, unlike Table 5, does not by itself separate population shift from top-k selection:

Table 6: Supporting object-decode distribution shift for the *off-policy-hardened* probe on CEM elite distribution (13,440 elites; final iteration = the converged population), vs. the same probe on random off-policy frames. The probe is more accurate on the random reference than on the selected CEM distribution; this unmatched comparison is supporting evidence only.

frame distribution	median err	within 5 cm	within 7 cm
random off-policy (reference)	2.0 cm	**91.5%**	—
all CEM elites	7.0 cm	24.0%	50.5%
final-iteration elites	7.3 cm	19.2%	46.0%
final-iteration, push	7.7 cm	**15.3%**	38.9%

The same probe that reads 2 cm on random frames reads 7–8 cm on selected elites. Because the random and CEM populations differ, these absolute numbers are not the primary selection test; the identical-population contrasts in Table 5 are. Simulator state nonetheless confirms that the selected objects are physically far from goal.

Simulator verification. Because ρ is itself OOD-sensitive, we verify the *same* elites against simulator state, which never passes through the probe: on the committed (final-iteration) elites the *true* object sits a median 24.9 cm from goal (19.9 cm on push, 28.0 cm on pick), and only 1.7% fall within the 5 cm success radius (push 3.4%, pick 0.0%). The planner's trusted candidates are genuinely far from the goal. Ground-truth elite quality (1.7% within 5 cm) is in fact *worse* than the probe reports (19.2%), so the probe number understates the wall (`results/exploit_simstate_crosscheck.csv`).

6.3 Shared-population branch intervention

Same-population regret establishes immediate misselection but not the consequence of adaptive refitting. We therefore fork CEM at the first population: both branches start from the identical simulator snapshot and candidate-for-candidate identical actions; one refits on stateprobe cost and the other on simulator-state cost. Later iterations use common Gaussian noise transformed by each branch's own mean and variance. All candidates continue to be rolled in MuJoCo and scored by both objectives.

Table 7: Shared-population branch intervention (n=8 unseen seeds). Immediate regret is the physical regret of the stateprobe argmin on the identical first population. Final Δ best physical cost is stateprobe-refit minus truth-refit at iteration 5. All CIs exclude zero.

cell	immediate physical regret	final Δ best physical cost
DINO push	1.71 cm $[1.22, 2.21]$	1.64 cm $[1.14, 2.20]$
DINO pick	1.95 cm $[1.33, 2.57]$	1.51 cm $[0.98, 2.06]$
JEPA push	1.44 cm $[0.97, 2.04]$	2.01 cm $[1.65, 2.35]$
JEPA pick	1.65 cm $[1.24, 2.05]$	1.77 cm $[0.96, 2.64]$

Thus the cost used for selection changes what the optimizer proposes later: after five refits, the stateprobe branch contains a worse *best* physical candidate in every tested cell. The branch difference in population decode error is CI-clean only for DINO push/pick, so we do not claim a universal monotonic decoder-error drift. The robust consequence is degradation in best simulator-state cost.

6.4 Coverage and ranking are simultaneous bottlenecks

An IMWM-style coverage audit asks whether an exactly successful candidate is present before blaming selection. At the final CEM iteration, exact-success coverage is only 8.0% of replans for DINO push, 3.6% for DINO pick, and 0% for both JEPA contact cells. Consequently, many failures cannot be repaired by reranking the existing population. Yet selected physical regret is positive and rank agreement remains low

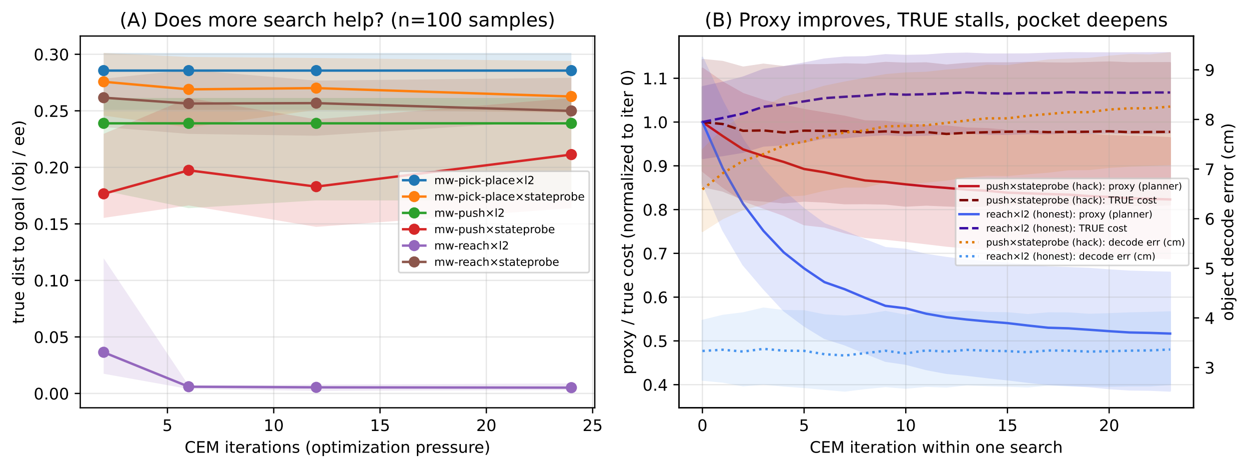

Figure 1: **Search pressure on latent planning costs** (n=16 paired episodes per cell; shaded bands are episode-clustered bootstrap 95% CIs). *(A)* True distance to goal vs. CEM iterations. reach×L_2 (informative cost) converts more search into success (5/16 → 16/16); push/pick×probe-cost (misranking) and push/pick×L_2 (no-signal) do not improve with budget. *(B)* Within one search at the largest budget, normalized to the first CEM iteration: for push×probe-cost the proxy the planner minimizes falls (−18%) while the true state cost stalls (−2%) and the object-decode error on the committed elites rises in point estimate (6.6 → 8.3 cm, +25%; pick 6.7 → 8.1 cm, +21%). The decode rise is a point-estimate trend (first/last CIs overlap at this n); the informative reach cost (reach×L_2) shows no such trend (+1%).

on the same candidates. We therefore retain a joint conclusion: proposal coverage limits contact planning, and the representation-derived cost additionally misranks available physical progress. Neither is claimed as the sole bottleneck.

6.5 Search-pressure curves

As supporting evidence, we sweep the CEM budget. An informative true-state cost should convert extra search into better true outcomes; a misleading cost may improve its proxy without comparable physical progress. We test this on the latent oracle (perfect dynamics; only the cost is learned, so any proxy/truth divergence is attributable to the cost under optimization, not to predictor error), sweeping CEM iterations $\in \{2, 6, 12, 24\}$ over n=16 paired episodes per cell, across cost×task arms ({push, pick-and-place, reach}×{probe-based state cost, latent L_2}). The relevant arms provide an informative positive control (reach×L_2), a misranking cost (contact×stateprobe), and a no-signal cost (contact×L_2).

The positive control and proxy–truth divergence are clear, while monotonic decode deepening remains suggestive (Fig. 1). **Probe cost** (push, probe-based state cost): within a single search at budget 24, the believed-best candidate's proxy falls monotonically (−18%) while its true object→goal distance does not improve (−2%), and the probe's decode error on the elite population rises 6.6 cm [5.7, 7.4] → 8.3 cm [7.1, 9.3] (+25%). We state the evidentiary status of that last quantity precisely: the first/last-iteration decode CIs *overlap* at n=16, so the increase is a point-estimate trend, not a CI-separated effect. Pick-and-place shows the same direction (6.7 → 8.1 cm, +21%), whereas the informative reach control changes only +1%. Across budgets the true push object distance does not improve with search (med ≈ 0.18–0.21 m at all of $\{2, 6, 12, 24\}$ iterations; episode success 0–2/16, every CI containing the frozen band). **Informative positive-control cost** (reach, L_2): here more search *does* convert to success (5/16 → 16/16 → 16/16 → 16/16 at 2/6/12/24 iterations), confirming the sweep detects "search helps" when the cost is aligned. **No-signal cost** (push, L_2): the proxy falls (−35%) but the true object distance is *exactly* flat (0.239 m median at every budget) — under this task and proposal distribution, optimizing L_2 is inert. Pick-and-place corroborates the pattern (probe-cost 0–1/16 with the same rising-decode signature; L_2 flat at 0.286 m). A second pressure axis — CEM population $\in \{50, 100, 300\}$ at fixed 12 iterations (n=8 episodes) — corroborates: enlarging the candidate pool does not improve the true push outcome either (probe-cost median 0.233/0.207/0.252 m; L_2

flat at 0.252 m).

6.6 Tested mitigation arms do not cross contact

We next test whether standard post-hoc repairs improve contact success, rather than inferring success from their average proxy metrics. We ran each arm under the latent oracle (perfect dynamics), on held-out push episodes with strict success:

Table 8: Mitigation ladder (latent oracle, push; 16 episodes per arm except the learned-metric arm at 8). Every intervention that operates on or above a frozen-base representation stays inside the frozen baseline's 0–2/16 band — including those that improve their object-only grounding metric. These metrics do not certify the full hand–object cost. The true-state row is the privileged positive control.

intervention	grounding-level metric	push
latent L_2 (frozen null)	—	0/16
probe-based state cost	static decode 92% <5 cm	2/16
off-policy-hardened probes	off-policy 91.5% <5 cm	1/16
learned latent metric d_θ	—	0/8
relearned repr. adapter φ (v2)	held-out decode med. 6.9 cm	1/16
encoder-LoRA + φ (5 seeds)	decode 94–100% <5 cm	{5,0,2,1,1}/16; mean 1.8, CI $[0, 3.7]$
+ ensemble disagreement, $\lambda \in \{0, \frac{1}{2}, 1, 2\}$	—	{2,1,2,1}/16
true-state cost (privileged control)	—	**16/16**

Three of these deserve comment. **Encoder fine-tuning** (LoRA on the DINOv2 trunk, action-conditioned grounding objective, 5 seeds): grounding reaches 94–100%, the five-seed planning mean is 1.8/16 with CI $[0, 3.7]$ — statistically inside the frozen band; the single 5/16 seed is high-variance noise, not a crossing (its own 95% Wilson interval, $[14, 56]$%, overlaps every other arm's; and every frozen-base arm's Wilson upper bound is $\leq 3.1/16$ for the n=16 rungs — `results/ladder_mitigation_cis.csv`). This is the strongest form of the paper's recurring scoped null: the tested grounding objective improves its target metric without producing reliable planning success. **Ensemble disagreement** (cost = $\text{mean}_k d_k^2 + \lambda \, \text{Var}_k[d_k]$ over 5 independently fine-tuned seeds) is the standard pessimism remedy for model exploitation (Janner et al., 2019; Coste et al., 2024); it fails here ($\{2, 1, 2, 1\}/16$ across λ) for an instructive reason: the seeds share the frozen pre-trained base, so they share the blind spot, and disagreement stays flat on the selected high-error candidates — the same failure mode reported for reward-model ensembles sharing a pre-trained trunk (Eisenstein et al., 2024). **The state-oracle row** is the positive control: the task and planner budget permit success when supplied true state, but it does not identify a unique defect shared by every failed learned cost.

Summary of the mechanism section. A frozen-base learned cost, under the tested search pressure, exhibits optimizer-conditioned misranking: CEM selects low-proxy candidates whose simulator-verified outcomes are worse than physical alternatives in the same population. The tested readout, grounding, metric, and disagreement repairs do not remove that failure. This is a finite scoped intervention grid, not evidence that every post-hoc cost must fail. Two caveats bound the strength of these nulls. First, statistical: at n=16 episodes a 0/16 arm's Wilson interval reaches 19.4%, so each null rules out mitigations that would lift success above ∼20%, not arbitrarily small gains — the claim is "no crossing," not "exactly zero." Second, multiplicity: the mitigation grid spans many small-n cells, which is precisely why we read the occasional 2/16 (and the single 5/16 seed) as band noise rather than signal — the same logic applies symmetrically and we do not interpret any individual +2/16 cell as a partial success.

7 Supporting predictor intervention: counterfactual training

The ladder bypassed the predictor (perfect dynamics); the deployed stack still uses one. Independently of the cost-side wall, the diagnostic of Sec. 3 identifies a training-target defect in the predictor: under a teacher-forced objective, factual prediction error does not directly require correct ranking across alternative actions. One possible corrective is to make the objective *counterfactual over actions*: the prediction under the factual action must beat predictions under distractor actions, not just match its target.

We instantiate this as an InfoNCE-over-actions objective on the frozen DROID world model (`dino_wm_droid`): a predictor-LoRA of 0.469M parameters (encoder untouched, no object ground truth), trained so that $F(z_t, a_t)$ scores higher against z_{t+1} than $F(z_t, a_k^-)$ for executed-action distractors a_k^-, with a stop-gradient on the target. Concretely, writing $d(a) = \|F(z_t, a) - \mathrm{sg}(z_{t+1})\|^2$ for the predictor's latent error under action a (sg the stop-gradient) and $\{a_k^-\}_{k=1}^K$ for in-batch permuted (executed) distractor actions, the objective is the action-softmax cross-entropy

$$\mathcal{L}_{\mathrm{cf}} \;=\; -\log \frac{\exp\big(-\,d(a_t)/\tau\big)}{\exp\big(-\,d(a_t)/\tau\big) + \sum_{k=1}^{K} \exp\big(-\,d(a_k^-)/\tau\big)}, \tag{4}$$

with temperature $\tau = \mathrm{sg}\big(\overline{d(a_t)}\big)$ auto-scaled to the batch-mean factual distance, so the logits stay $O(1)$ regardless of the model's absolute latent scale (no per-model tuning). The stop-gradient on z_{t+1} is essential: without it the objective is minimized by collapsing the *target* rather than by grounding the *prediction*. An earlier version of this draft reported large gains from this objective on a DROID planning probe. That evaluation was invalid as a held-out claim: although training used trajectory-level train/validation partitions, the downstream planning probe could draw anchors and hard negatives from the complete latent cache, including training trajectories. We therefore remove those numerical results and all method claims based on them.

The corrected protocol persists an immutable 70/15/15 trajectory manifest, embeds its membership and hash in each checkpoint, selects checkpoints on validation only, and restricts both planning anchors and hard-negative pools to the untouched test trajectories. Across four training seeds, DINO-WM improves the offline DROID Action Error ($1.492 \to 1.085$), Action Score ($0.471 \to 0.545$), and CRA ($0.036 \to 0.233$). JEPA-WM improves CRA ($0.030 \to 0.085$) but not Action Error ($1.337 \to 1.344$) or Action Score ($0.501 \to 0.501$). These mixed, offline results are supporting evidence only: they are not task success and do not establish a method contribution. The oracle ladder also implies that even a genuine predictor-side gain would not, by itself, remedy the cost-side failure, because learned dynamics have already been removed in the latent-oracle arms.

8 Discussion, scope, and limitations

Optimizer-conditioned audits. The primary audit uses one candidate population and two scorers: proxy $\hat{c}$ and simulator truth c^*. It reports (i) rank correlation and top-k overlap, (ii) opportunity regret R_{sel}, (iii) optimism enrichment

$$\Delta_{\mathrm{opt}} = \mathbb{E}[c^* - \hat{c} \mid \text{proxy elite}] - \mathbb{E}[c^* - \hat{c} \mid \text{population}], \tag{5}$$

and (iv) whether an exactly successful candidate was covered before selection. These answer distinct questions. Positive regret identifies misranking among available candidates; low coverage identifies a proposal limitation; neither alone predicts closed-loop success. Random-off-policy-to-elite readout shift is reported only as supporting distribution-shift evidence because those populations are not matched.

The shared-branch intervention adds the adaptive test: hold the initial population and common noise fixed, vary the cost used for refitting, and compare the best simulator-state cost subsequently available. This turns the audit from a post-hoc correlation into an intervention on the planner–cost interface. The result remains a mechanism diagnostic rather than a universal calibrated metric; task success is the endpoint.

What follows for method design. The evidence motivates three non-exclusive responses: learn a cost whose *candidate ordering* is validated under adaptive search; improve proposal coverage with demonstrations or action priors; and train task-relevant relative state rather than optimizing an object-only decoding metric. The tested post-hoc repairs in Table 8 do not cross contact, but this is a finite null grid, not an impossibility result. TRM- and ACID-style comparisons are still running and will determine whether an existing reachability or consistency cost repairs the tested harness.

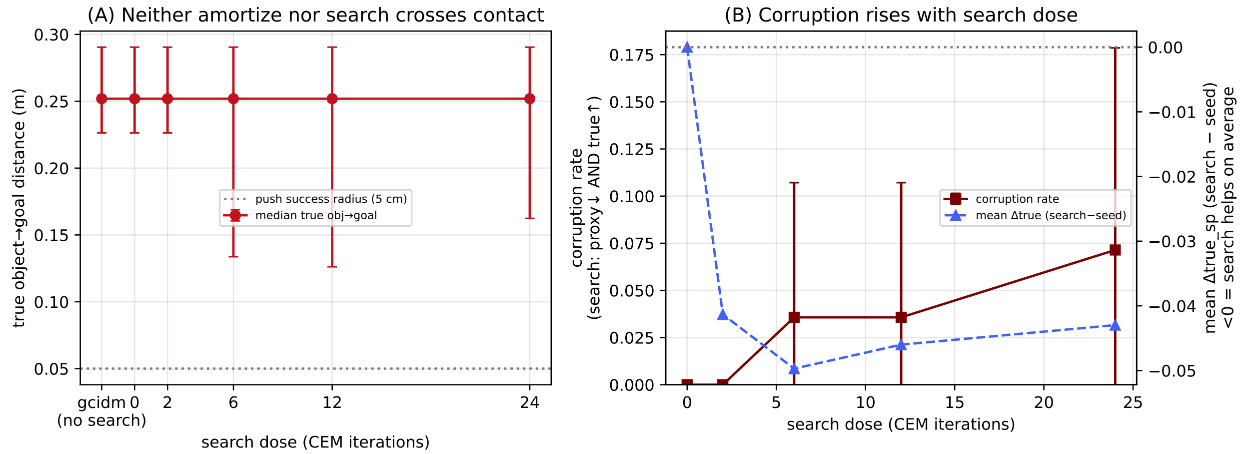

Figure 2: **Removing search does not cross contact for this controller.** *(A)* True object→goal distance for the pure amortized controller ("gcidm", no search) and for search re-added in doses; all sit at ≈ 0.25 m, far above the 5 cm success radius. *(B)* Per-replan corruption rate (search lowers the proxy *and* raises the true cost) rises with search dose ($0 \to 7\%$), while the mean true-cost change from search stays negative (on average search still helps the *hand*) — the measured selection effect is small here, and the object never moves regardless.

Search-free controller control. Under the latent oracle on push, the search-free amortized controller solves 0/8, and every re-added search dose (CEM iterations $\{0, 2, 6, 12, 24\}$ seeded from the controller) also solves 0/8; in all arms the object never leaves its start (med $\|obj - goal\| \approx 0.25$ m, Fig. 2A). Removing the optimizer does not cross the wall. Two things are nonetheless confirmed. On the mechanism side, the per-replan seed-vs-chosen comparison shows search "improves" the proxy over the seed on 98–100% of replans while *corrupting* the true state (proxy down, truth up) on a fraction that rises monotonically with dose ($0 \to 7\%$, Fig. 2B) — a small optimizer-conditioned effect in this control. But it is not the only observed failure: this particular controller also fails without an optimizer. That null does not identify whether representation information, controller training, coverage, or control architecture is the cause. The supported synthesis is narrower: the tested search/cost interface exhibits optimizer-conditioned misranking, and neither the tested cost repairs nor this amortized controller crosses contact on its own. Grounding metrics are therefore insufficient as success certificates; their necessity remains unestablished.

Scope and honest limits. The cost-misranking mechanism is established on MetaWorld, where simulator state lets us verify candidate ordering against ground truth; on DROID the evidence remains an observational action-discriminability diagnostic. The shared-mask scaling rerun remains pending; the held-out predictor intervention is mixed (DINO improves offline Action Error/Score, JEPA does not) and is not a closed-loop result. Closed-loop transfer to real robots remains future work. The mitigation table covers the levers available on frozen pretrained checkpoints (readouts, metrics, adapters, LoRA fine-tuning, ensembles); full encoder retraining with planner-aware objectives is beyond our compute and remains open — our claim is about the *post-hoc* regime. The boundary-regime labels use an object-displacement proxy (no MuJoCo contact ground truth in the released data); mw-door-close is excluded as a proxy anomaly. The Reach parity check compares one released checkpoint over 16 episodes against a published 3-seed$\times$96-episode number — we read it as "reproduces, inside the CI", not a seed-matched replication. Finally, the current probes do not distinguish information absent from the encoder from information inaccessible to the readout under search. The state-oracle row shows the *task* is solvable from privileged state, but whether a robust vision-only cost can be learned is not settled by this paper.

9 Conclusion

On the tested MetaWorld contact tasks, planning failure persists when learned dynamics are replaced by simulator-perfect rollouts, while the same finite planner succeeds with privileged state cost. On identical populations, the probe cost selects candidates with optimistic residuals and positive physical regret; changing only the refitting cost subsequently changes the best physical candidate available. Exact success coverage is also sparse, so the diagnosis is a joint proposal-and-ranking bottleneck. The result is not a universal impossibility claim about encoders, latent costs, scale, or grounding. It is a controlled argument that world-model planners should report physical candidate ordering, opportunity regret, and coverage under the optimizer's own proposal distribution, not object-only decoding or aggregate prediction metrics alone.

References

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.

Mahmoud Assran et al. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.

Jiaheng Chen et al. ATM: Action-consistency transfer matrix for diagnosing and improving latent world models. *arXiv preprint arXiv:2606.09028*, 2026.

Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *NeurIPS*, 2018.

Thomas Coste, Usman Anwar, Robert Kirk, and David Krueger. Reward model ensembles help mitigate overoptimization. In *ICLR*, 2024.

Jacob Eisenstein, Chirag Nagpal, Alekh Agarwal, Ahmad Beirami, Alex D'Amour, DJ Dvijotham, Adam Fisch, Katherine Heller, Stephen Pfohl, Deepak Ramachandran, Peter Shaw, and Jonathan Berant. Helping or herding? reward model ensembles mitigate but do not eliminate reward hacking. In *Conference on Language Modeling (COLM)*, 2024.

Baoqi Gao, Ruize Han, Miao Wang, and Song Wang. IMWM: Intuition models complement world models for latent planning. *arXiv preprint arXiv:2606.01626*, 2026.

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *ICML*, 2023.

Adam Gleave, Michael Dennis, Cody Wild, Neel Kant, Sergey Levine, and Stuart Russell. Adversarial policies: Attacking deep reinforcement learning. In *ICLR*, 2020.

Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *NeurIPS*, 2019.

Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. Goodhart's law in reinforcement learning. In *ICLR*, 2024.

Alexander Khazatsky et al. Droid: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024.

Rahul Kidambi, Aravind Rajeswaran, Praveen Palanisamy, and Sham Kakade. MOReL: Model-based offline reinforcement learning. In *NeurIPS*, 2020.

Nathan Lambert, Brandon Amos, Omry Yadan, and Roberto Calandra. Objective mismatch in model-based reinforcement learning. In *L4DC*, 2020.

Liangyu Li, Shengzhi Wang, and Qingwen Liu. Beyond Euclidean proximity: Repairing latent world models with horizon-matched trajectory reachability metrics. *arXiv preprint arXiv:2605.22164*, 2026a.

Wenyuan Li, Guang Li, Keisuke Maeda, Takahiro Ogawa, and Miki Haseyama. Predictive but not plannable: RC-aux for latent world models. *arXiv preprint arXiv:2605.07278*, 2026b.

Yuejiang Liu, Fan Feng, Lingjing Kong, Weifeng Lu, Jinzhou Tang, Kun Zhang, Kevin Murphy, Chelsea Finn, and Yilun Du. World action verifier: Self-improving world models via forward-inverse asymmetry. *arXiv preprint arXiv:2604.01985*, 2026.

Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.

Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar, Mido Assran, Koustuv Sinha, Mike Rabbat, Yann LeCun, Nicolas Ballas, and Adrien Bardes. V-JEPA 2.1: Unlocking dense features in video self-supervised learning. *arXiv preprint arXiv:2603.14482*, 2026.

Anusha Nagabandi, Kurt Konolige, Sergey Levine, and Vikash Kumar. Deep dynamics models for learning dexterous manipulation. In *CoRL*, 2019.

Hoang Nguyen, Xiaohao Xu, and Xiaonan Huang. Latent geometry beyond search: Amortizing planning in world models. *arXiv preprint arXiv:2605.08732*, 2026.

Arjun Parthasarathy, Nimit Kalra, Rohun Agrawal, Yann LeCun, Oumayma Bounou, Pavel Izmailov, and Micah Goldblum. Closing the train-test gap in world models for gradient-based planning. *arXiv preprint arXiv:2512.09929*, 2025.

Reuven Y. Rubinstein. Optimization of computer simulation models with rare events. *European Journal of Operational Research*, 99(1):89–112, 1997.

Gawon Seo, Dongwon Kim, and Suha Kwak. ACID: Action consistency via inverse dynamics for planning with world models. *arXiv preprint arXiv:2607.02403*, 2026.

Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *NeurIPS*, 2022.

Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models? *arXiv preprint arXiv:2512.24497*, 2025.

Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *CoRL*, 2020a.

Tianhe Yu, Garrett Thomas, Lantao Yu, Stefano Ermon, James Zou, Sergey Levine, Chelsea Finn, and Tengyu Ma. MOPO: Model-based offline policy optimization. In *NeurIPS*, 2020b.

Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. Dino-wm: World models on pre-trained visual features enable zero-shot planning. *arXiv preprint arXiv:2411.04983*, 2024.

A Regime stratification

Every per-regime table in the paper (Tables 1, 2, and 9) relies on assigning each transition $(s_t \to s_{t+1})$ to exactly one of four regimes. The assignment is a *priority cascade*: the first matching rule wins, so the regimes are mutually exclusive by construction and the ordering encodes precedence (the gripper-actuation *event* outranks the contact *state* it may create). Thresholds are fixed constants (below), not fit to any outcome.

Metaworld (simulator state). The released HuggingFace dataset ships no MuJoCo contact flags but does ship the full 39-dim Metaworld-v2 observation, from which we recover the end-effector position (`state[0:3]`), gripper aperture (`state[3]`), and primary-object position (`state[4:7]`) exactly. For a transition:

1. **gripper actuation** if $|\Delta\text{grip}| > 0.10$ (the gripper is opening/closing — the grasp/release event itself);
2. else **contact manipulation** if $\|\Delta\text{obj}\| > 5\,\text{mm}$ (the object measurably moved — something pushed it);
3. else **pre-grasp** if $\|\text{ee} - \text{obj}\| < 10\,\text{cm}$ (the hand is close to the object but has not yet moved it);
4. else **free space**.

"Contact" here therefore means *the object displaced*, an object-displacement *proxy* rather than a contact sensor; the caveat is carried on every Metaworld figure, and `mw-door-close` is excluded as a proxy anomaly (its hinge geometry violates the displacement assumption).

DROID (real gripper sensor + latent-change proxy). DROID-wrist has no object pose, so contact is read from the *real gripper sensor* (primary) and the approach cue from the latent-change proxy (secondary):

1. **gripper actuation** if $|\Delta\text{grip}| > 0.2$;
2. else **contact manipulation** if the gripper is held *closed* ($\text{grip}_t > 0.5$ on the $[0, 1]$ open→closed scale) — holding or contacting an object, a sensor signal that (deliberately) does *not* depend on visual motion, so a closed gripper resting on an object counts as contact even when the scene is momentarily static;
3. else (gripper open) **pre-grasp** if the scene is actively changing, $\|\Delta z\| > 1.0 \times \text{median}\|\Delta z\|$ (an object entering/growing in the wrist view; the dataset-median baseline is the same one used for effect-conditioning);
4. else **free space** (gripper open, scene static).

Making contact gripper-primary rather than $\|\Delta z\|$-gated matters for the raw per-regime counts (the earlier motion-gated rule mislabeled ∼54% of closed-gripper *static-holding* steps as free space); it leaves the *headline* numbers unchanged, because those are effect-conditioned ($\|\Delta z\| >$ median) and the relabeled steps are the low-$\|\Delta z\|$ holding steps that effect-conditioning excludes either way.

RoboCasa (grasp flags). RoboCasa exposes an `object_in_hand` flag and proprioception, mapped to the same taxonomy: gripper actuation if $|\Delta\text{grip}| > 0.2$; else contact manipulation if `object_in_hand` at t or $t+1$; else pre-grasp if $\|\text{ee} - \text{obj}\| < 12\,\text{cm}$ (looser than Metaworld for the larger scene scale); else free space.

The three datasets thus share one taxonomy and one cascade ordering, differing only in the contact signal available (object displacement / gripper sensor / grasp flag). Implementations are in the dataset-specific modules under `diagnosis/stratification/`.

Robustness to the threshold choice. Because the pre-grasp locus (Table 1) is read off a fixed 5 mm/10 cm/0.10 cut, we re-ran classification and the hard-nearest-neighbour diagnostic (Sec. 3.1) under six perturbations — object-move $\in \{2.5, 10\}$ mm, pre-grasp distance $\in \{8, 12\}$ cm, gripper-delta $\in \{0.05, 0.20\}$ — against the baseline. Pooled effect-conditioned CRA at pre-grasp moves by at most 0.037 (DINO-WM: $[0.450, 0.488]$; JEPA-WM: $[0.529, 0.558]$) across all seven configurations, and **pre-grasp remains the lowest-scoring regime, below both free space and contact manipulation, in every single configuration for both models** (`results/regime_robustness_summary.csv`, scripts/04+46). The locus of the collapse is a property of the transitions, not an artifact of the specific cut.

B Boundary Blindness: definition

The instrument compares, on a neighbourhood of similar states, how much the *world's* outcome fans out under nearby actions against how much the *model's* prediction fans out under the same actions.

Neighbourhood. For each anchor transition i with latent z_i and action a_i, $\mathcal{N}(i)$ is its $M{=}16$ nearest cached transitions in latent distance, radius-limited with relaxation (the same neighbourhood the hard-negative CRA sampler uses). o_j is the true scalar outcome of neighbour j: object displacement $\|\Delta \mathrm{obj}_j\|$ on Metaworld, the $\|\Delta z_j\|$ proxy on DROID.

Boundary regime. A transition is *boundary* when small action changes produce large true-outcome changes across its neighbourhood:

$$\text{boundary\_score}_i = \frac{\mathrm{std}_j \ o_j}{\mathrm{mean}_j \ \|a_j - a_i\| + \varepsilon}, \tag{6}$$

thresholded at the 75th population percentile. Free-space drift has large $\|\Delta z\|$ but smooth neighbourhoods and scores low.

The two sensitivities and BB. The world's local action-sensitivity S_i^{true} is the dispersion of true outcomes across the neighbourhood; the model's, S_i^{model}, applies each neighbour's action to the *single* anchor state and measures the RMS spread of $F(z_i, a_j)$ around its centroid. Each is z-scored over the population and

$$\mathrm{BB}_i = \mathrm{relu}\left(\tilde{S}_i^{\text{true}} - \tilde{S}_i^{\text{model}}\right), \tag{7}$$

pooled (n_b-weighted) over the boundary mask with trajectory-clustered bootstrap CIs. Because BB references the world, a high value cannot be an artefact of small one-step latent deltas.

C The grounded-channel fix ladder (metric-level repair)

Deferred from Sec. 4: the model-side repair ladder that motivated the oracle ladder. **Nulls:** mixture-density predictor heads (soft-NLL/WTA/hard-EM/supervised, $K{\in}\{2,3\}$) beat their $K{=}1$ NLL controls but do not move $\mathrm{BB}_{\text{boundary}}$ — the bifurcation spans 9.9 L_2 units against a 106 median residual (the geometry under-weights the boundary subspace 10×), and the boundary's action-dependence is unlearnable from expert-only data at the head level. Probe-subspace metric re-weighting redistributes BB without reducing it. **The probe chain:** object state is decodable from the frozen latent (V1 err 0.064 vs. sd 0.094), propagates through the predictor for *factual* actions (V2 0.059), but the *counterfactual* action→object response is noise (V3 spread-corr +0.035). **The repair:** a 0.5M grounded dynamics channel $h(z_t, a) \to \widehat{\Delta \mathrm{obj}}$ (mean/max-pooled tokens + action MLP), trained on cached latents with everything frozen, restores counterfactual tracking (0.035→0.682 DINO-WM; 0.702 JEPA-WM), halves boundary BB on both families (1.323→0.660; 1.280 → 0.620), and improves the closed-loop cost surface (Table 3) — while flipping zero successes, the observation Sec. 5 explains. Honest negative: the recipe does not transfer to DROID's noisy whole-state proxy label at 2.1k transitions (rank-corr +0.708 but spread magnitude collapsed; BB unchanged) — the principle needs a learnable grounding label.

D Probe, verifier, and mitigation-arm implementations

Every learned rung of the oracle ladder (Table 4) and mitigation table (Table 8) is small and trained on the cached latents with the encoder and predictor frozen. We give each arm's spec so the "all repairs fail" claim can be audited.

Object probes (the readout ρ). Two readouts decode object xyz from a single frozen latent, trained supervised on cached (latent, simulator object-position) pairs. The *pooled* probe is a mean+max-pooled-token MLP ($2D{\to}512{\to}256{\to}3$, SiLU). The *spatial* probe prepends a learned CLS query to the patch tokens and routes it through a 2-layer, 6-head Transformer encoder before a linear head; being permutation-aware it can attend to *which* patch lit up, so it tests accessibility under a stronger nonlinear readout than global pooling — it is the readout behind the 92%-within-5 cm decode in Table 4. The *expert* probe trains on expert-trajectory frames only; the *off-policy-hardened* probe mixes in latents of arbitrary candidate rollouts, which restores 91.5% within 5 cm on that off-policy distribution.

On using ρ as the verifier. The unmatched distribution-shift table (Table 6) decodes ρ on CEM elites, but ρ is itself OOD-sensitive. We therefore do not use it as ground truth. Every candidate in the primary same-population, permutation-null, coverage, and shared-branch audits is rolled in MuJoCo and scored independently from simulator state (Sec. 6.1). The probe supplies the deployed proxy; simulator state supplies physical ordering and regret. On real-robot data such privileged verification is unavailable, which is why DROID remains observational supporting evidence rather than a causal replication.

Grounded object-dynamics channel h (Sec. 4, App. C). $h(z_t, a) \to \widehat{\Delta\mathrm{obj}}$ is a mean+max-pooled-token MLP with an action-embedding branch ($2D{+}64 \to 512 \to 256 \to 3$); the grounded cost integrates it along the rollout, $\widehat{\mathrm{obj}}_H = \rho(z_0) + \sum_t h(\hat{z}_t, a_t)$, and scores $\mathrm{MSE}(z_H, z_g)/s_z^2 + \beta^2 \|\widehat{\mathrm{obj}}_H - \rho(z_g)\|^2/s_g^2$, the two terms rescaled by dataset scales s_z, s_g because they span ∼98k vs. 3 dimensions (per-dim MSE would silently collapse the blend to object-only).

Learned latent metric d_θ (Tables 4, 8). A goal-conditioned quasimetric over pooled tokens, $d_\theta(z, z_g) = \|u_s(z) - u_s(z_g)\|_2 + \sum_i \mathrm{relu}\big(u_a(z)_i - u_a(z_g)_i\big)$ — a symmetric part plus a directed residual (MRN-style), so $d_\theta(z_g, z_g) = 0$ is the unique minimum. Trained *without* object labels on within-trajectory temporal order ($d_\theta(z_t, z_{t'}) \approx t' {-} t$ for $t < t'$, cross-trajectory pairs pushed large), so the recipe is dataset-agnostic.

Relearned representation adapter φ (Table 8). $\varphi = A(z)$ is a small head over the frozen patch tokens, trained with four losses so that plain L_2 in φ-space is a harder-to-exploit cost: (i) *grounding* — the first obj_dim coordinates regress true object xyz; (ii) *counterfactual margin* — on hard-effect negatives (same z_t, different action, genuinely different true outcome) the two object estimates must be separated by at least their true gap, a margin constraint that targets residual-error *pockets* rather than average accuracy; (iii) *temporal ranking* confined to the non-object subspace; (iv) *adversarial* — previously-mined CEM elite frames are margin-pushed from their episode goal in the object subspace. This is the strongest post-hoc cost we built, and it still gates push 1/16.

Encoder-LoRA (Table 8). Zero-initialized LoRA adapters ($B{=}0$, so the injected model starts *exactly* at the frozen encoder) inserted into the encoder's transformer blocks and trained with the φ losses; unlike φ — a function of the frozen z, so it can only read around entangled latents — this can separate them at the source. The adapters are toggleable, so one injected model serves both the frozen-baseline and LoRA arms; we train five seeds.

Ensemble disagreement (Table 8). Five independently fine-tuned encoder-LoRA seeds, with cost $\mathrm{mean}_k d_k^2 + \lambda\, \mathrm{Var}_k[d_k]$ — the standard pessimism penalty — swept over $\lambda \in \{0, \frac{1}{2}, 1, 2\}$.

E Reproduction integrity

Closed-loop sim evaluation is fragile, and getting the baseline wrong would invalidate the parity argument. Three environment-side bugs, found and fixed, all model-agnostic: **(1)** Metaworld's offscreen renderer is constructed in the env constructor, so camera/size assignments afterwards were silently ignored (default camera at 480 px vs. the trained `corner2` at 224 px; symptom: one-step prediction error 8.5× dataset level). We re-create the renderer explicitly, mirroring the upstream wrapper. **(2)** The released training frames are vertically flipped relative to today's `corner2` render; pixel-calibration against dataset initial frames scores the flipped wrapper-camera render at MSE 71.6 vs. ≥ 3000 for every unflipped candidate. **(3)** The success flag fires at a 5 cm radius but the scripted expert refines to ∼1 cm; taking the goal frame at first-success rather than the expert's final frame parked the planner just outside the radius. **Verification:** after the fixes, one-step prediction error is 1.4× dataset level, latent NN ratio 0.97, and replaying dataset actions reproduces the proprioceptive trajectory to 1.5 mm median end-effector error — only the visuals had drifted, never the physics. The reach baseline then recovers from near-zero to 37.5% (Table 3), inside the published CI.

F AUG and ECS: the raw-gap corroboration

CRA leads the main text because, as a ranking, it is cross-model comparable. The two raw-gap metrics corroborate it: AUG (prediction-error gap between permuted and factual actions; ≈ 0 iff actions ignored) and ECS (AUG on effect-bearing transitions), both on the hard nearest-neighbour family.

Table 9: AUG / ECS, hard nearest-neighbour family, per regime (n_{eff}-weighted pools; dash = no effect-bearing transitions). Raw latent gaps are *not* comparable across models (latent scales differ); the cross-model statement lives in CRA (Tables 1, 2).

model	free space	pre-grasp	gripper	contact
DROID (real robot; AUG / ECS)				
DINO-WM (22M)	0.038/–	0.011/0.0105	0.038/0.0645	0.070/0.0094
JEPA-WM (300M)	0.000/–	0.000/0.0002	0.001/0.0004	0.001/−0.0011
V-JEPA-2-AC (1B)	0.006/–	0.002/0.0016	0.003/0.0026	0.003/−0.0059
V-JEPA-2-AC-OSS (1B)	0.004/–	0.001/0.0010	0.001/0.0005	0.003/−0.0030
Metaworld (sim; AUG / ECS; pooled over 12 tasks)				
DINO-WM (22M)	0.088/0.1007	0.051/0.0997	–/–	0.081/0.0893
JEPA-WM (300M)	0.089/0.1020	0.058/0.1114	–/–	0.095/0.1052

Under the model-native DROID protocol, the effect-conditioned action gap remains small across the released checkpoints. This observational result is compatible with weak action use, but it does not by itself identify causal action-ignoring or constitute a controlled scale law. Source: the `hard_nn` rows in the DROID and MetaWorld diagnostic CSVs under `diagnosis/results/`.

G Full per-task tables

Per-task Boundary-Blindness dumps, per-arm mitigation episode tables (phase C/D/F/G logs), the overoptimization-sweep curve tables, and the historical and strict-held-out Phase-H seed-sweep outputs are provided in machine-readable form under `diagnosis/results/`. Historical overlapping splits are retained only for provenance; current claim status is mapped in `diagnosis/docs/CLAIMS_EVIDENCE.md`.